Miraslau Pisaryk

Java / Spring Boot Backend Engineer | BSUIR Student & Production Portfolio

bhasdgjkg5@gmail.com | +375 29 749-77-19 | Telegram: @yxxtg | Minsk, Belarus
github.com/CHIKOJgg | linkedin.com/in/miroslav-pisaryk-953490261 | **Open to Internships / Junior Positions / Remote Work**

SUMMARY

Computer Science student at BSUIR specializing in building production-grade distributed backends on Java 21/25 and Spring Boot 3/4. Practical experience: thread-safe in-memory matching engine, real-time WebSocket streaming, event-driven microservices with Spring AI RAG and autonomous Function Calling, Keycloak 26 OAuth2 IAM integration. 100% Testcontainers test isolation. English C1 Advanced.

SKILLS

Languages & Core: Java (17, 21, 25), SQL, Concurrency (Virtual Threads / Loom, ReentrantLock, CompletableFuture, ConcurrentHashMap), Stream API, Generics, OOP/SOLID

Frameworks: Spring Boot 3/4, Spring AI 1.0, Spring Cloud OpenFeign, Spring Data JPA / Hibernate, Spring Security (OAuth2 Resource Server), WebSocket (STOMP), Resilience4j, Bucket4j, MapStruct

IAM & Security: Keycloak 26 (OIDC, OAuth2 Authorization Code + PKCE, JWT JWKS Verification, Realm Role Mapping, JIT User Provisioning, TokenRelayInterceptor)

Messaging & Streaming: RabbitMQ 3.13 (Topic Exchanges, Dead-Letter Queues, AMQP, JSON Message Converter, Asynchronous Domain Events)

Databases & Search: PostgreSQL 17, pgvector (1536-dim Vector Embeddings, Cosine Similarity, HNSW Index), Redis 7 (Caching, Rate Limiting, Pub/Sub), Flyway Migrations

AI & LLM Engineering: Spring AI (ChatClient, @Tool Function Calling, VectorStore RAG, Dynamic System Prompts), OpenAI API, Model Context Protocol (MCP)

DevOps & Quality: Docker, Docker Compose, GitHub Actions CI/CD, Maven Multi-Module, Linux/Bash, JUnit 5, AssertJ, Mockito, Testcontainers (PostgreSQL, Redis, RabbitMQ)

Architecture: Event-Driven Microservices, Multi-Module Monorepo, DDD, Clean Architecture, REST API (RFC 7807 Problem Details), Agile Scrum/Kanban

Languages: English — C1 Advanced (EF SET 70/100, verified cert.efset.org/ru/cAsbkU); Polish — beginner

FEATURED ENGINEERING PROJECTS

AI Assistant Klawa — Distributed Event-Driven Microservices Architecture

Java 21, Spring Boot 3.4, Spring AI 1.0, PostgreSQL 17 + pgvector, Keycloak 26, RabbitMQ 3.13, OpenFeign, Redis 7, Testcontainers
github.com/CHIKOJgg/klawa | Maven Multi-Module Monorepo (contracts, core, ai-agent)

- **Monolith → Microservices Migration:** Architected the end-to-end migration from a Spring Boot monolith to an Event-Driven Microservices platform structured as a Maven multi-module monorepo with shared contracts (klawa-contracts, klawa-core, klawa-ai-agent).
- **Core Service & Virtual Threads:** Developed the high-throughput klawa-core service on Java 21 utilizing Virtual Threads (Project Loom) for non-blocking I/O across task orchestration, project management, Telegram adapter, and reminder scheduling.
- **Keycloak 26 Enterprise Security:** Integrated Keycloak 26 as an OAuth2 Resource Server with stateless JWT signature verification via JWKS endpoint, realm role mapping through custom JwtGrantedAuthoritiesConverter, and Just-In-Time (JIT) user provisioning.
- **Spring AI & pgvector RAG Pipeline:** Built the klawa-ai-agent microservice on Spring AI 1.0; implemented Retrieval-Augmented Generation using PostgreSQL 17 with pgvector (1536-dim HNSW index, cosine similarity) for real-time semantic context search over user tasks.
- **Autonomous @Tool Function Calling:** Implemented @Tool-annotated methods allowing the LLM to autonomously create tasks, change statuses, and query projects — calls execute synchronously in klawa-core via Spring Cloud OpenFeign with TokenRelayInterceptor (JWT propagation).
- **RabbitMQ Domain Event Bus:** Configured RabbitMQ 3.13 topology (Topic Exchange klawa.events, JSON MessageConverter) for asynchronous domain events (TaskCreatedEvent, AiTaskAnalysisRequestedEvent); decoupled embedding generation keeping HTTP latency <50ms.

- **Resilience & Observability:** Integrated Resilience4j Circuit Breakers for AI API failover, Bucket4j token-bucket rate limiting, Redis response caching, and 100% isolated integration tests via Testcontainers (PostgreSQL, Redis, RabbitMQ).

High-Frequency Trading Platform — In-Memory Matching Engine & Real-Time Streaming

Java 21, Spring Boot 3.3, PostgreSQL, Redis, WebSocket (STOMP), Spring Security (JWT), Flyway, Docker, GitHub Actions

- **FinTech Platform Core:** Developing the backend of a multi-currency banking and exchange platform for account management, trading, and real-time market quotes in a distributed Agile team of 8 engineers.
- **In-Memory Matching Engine:** Architected a thread-safe in-memory order matching engine with price-time priority for limit/market orders, leveraging Java Concurrency lock-free primitives (CAS, ConcurrentSkipListMap) to sustain 5,000+ ops/sec with zero data races.
- **Redis Caching & Load Reduction:** Integrated Redis for order book depth caching and message buffering, reducing PostgreSQL transactional read load by ~40% during peak market volatility.
- **Real-Time WebSocket Streaming:** Engineered real-time order book depth and private account notification streaming over WebSocket (STOMP) for 200+ concurrent clients with sub-100ms delivery latency.
- **Financial Schema & Migrations:** Designed a normalized 13-table PostgreSQL schema with double-entry accounting ledgers and immutable audit trails; automated multi-stage schema evolution via Flyway.

ADDITIONAL PROJECTS

Crypto Portfolio Optimizer — Quantitative Analysis & Backtesting Engine

Java 21, JavaFX, ojalgo, CoinGecko API | github.com/CHIKOJgg/MarkowitzModel-java-crypto-analysis

- **Quantitative Alpha Models:** Engineered 12 quantitative alpha strategies (Momentum, EWMA, RSI, MACD, Bollinger Bands) and 9 portfolio allocation frameworks (Markowitz Mean-Variance, Risk Parity, HRP, CVaR, Black-Litterman).
- **Backtesting & Risk Engine:** Built a walk-forward backtesting engine with Monte Carlo simulation (Geometric Brownian Motion) and real-time risk metrics: Sharpe, Sortino, Calmar ratios, VaR/CVaR 95%, Brinson attribution.

EDUCATION & CREDENTIALS

Belarusian State University of Informatics and Radioelectronics (BSUIR)

Bachelor's in Information Systems and Technologies (Computer Science) | 2024 – 2028 | Minsk, Belarus

English Certificate: EF SET — Score 70/100, C1 Advanced | cert.efset.org/ru/cAsbkU